STEP 1 — Install and Import Required Libraries

```
# Install required libraries
!pip install transformers
!pip install torch
```

```
lready satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
lready satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.2)
lready satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.6.0)
lready satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
lready satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
lready satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
lready satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
lready satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
lready satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
lready satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
lready satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
lready satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (2
lready satisfied: hf-xet<2.0.0,>=1.3.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers
lready satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0
lready satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformers) (0.24.1)
lready satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transfor
lready satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transform
lready satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transfo
lready satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->t
lready satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.3.0->transforme
lready satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0
lready satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers)
lready satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transfo
lready satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->transformers)
lready satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.0->trans
lready satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2.0
lready satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->huggingface-hub<2
lready satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->hugging
lready satisfied: torch in /usr/local/lib/python3.12/dist-packages (2.10.0+cpu)
lready satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from torch) (3.25.2)
lready satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
lready satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
lready satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
lready satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
lready satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
lready satisfied: fsspec>=0.8.5 in /usr/local/lib/python3.12/dist-packages (from torch) (2025.3.0)
lready satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.0)
lready satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
```

```
# Import required libraries
from transformers import pipeline
import torch

print("Libraries imported successfully!")
```

STEP 2 — Load Summarization Model

```
import torch
from transformers import AutoTokenizer, AutoModelForSeq2SeqLM

# Load summarization model and tokenizer directly
tokenizer = AutoTokenizer.from_pretrained("facebook/bart-large-cnn")
model = AutoModelForSeq2SeqLM.from_pretrained("facebook/bart-large-cnn")

print("Summarization model and tokenizer loaded successfully!")
```

```
vocab.json:        899k/?  [00:00<00:00,  7.75MB/s]

merges.txt:        456k/?  [00:00<00:00,  10.1MB/s]

tokenizer.json:        1.36M/?  [00:00<00:00,  16.1MB/s]

model.safetensors:  100%                                                    1.63G/1.63G  [00:26<00:00,  70.4MB/s]
Please make sure the generation config includes `forced_bos_token_id=0`.
Loading weights:  100%                                  511/511  [00:01<00:00,  425.52it/s,  Materializing  param=model.encoder.layers.11.self_attn_layer_norm.weight]

generation_config.json:  100%                                                    363/363  [00:00<00:00,  10.6kB/s]
Summarization model and tokenizer loaded successfully!
```

STEP 3 — Prepare Input Text Dataset

```
# Dataset texts (5–8 passages from different domains)

text1 = """
Artificial Intelligence is transforming industries worldwide. Companies are increasingly adopting AI technologies
to automate processes, improve customer service, and enhance decision-making. Machine learning algorithms can
analyze massive amounts of data faster than humans, identifying patterns that were previously difficult to detect.
AI applications range from healthcare diagnostics to financial fraud detection and autonomous vehicles.
However, concerns regarding data privacy, ethical use of AI, and potential job displacement remain significant
challenges for policymakers and organizations.
"""

text2 = """
Climate change is one of the most critical challenges facing humanity today. Rising global temperatures are causing
melting glaciers, sea level rise, and extreme weather events. Scientists warn that without significant reductions
in greenhouse gas emissions, the effects will become more severe. Governments around the world are working toward
sustainable solutions such as renewable energy adoption, carbon reduction policies, and environmental conservation.
Public awareness and global cooperation are essential to mitigate climate risks.
"""

text3 = """
In software development, version control systems are essential tools for managing code changes. Git is one of the
most widely used systems that allows developers to track modifications, collaborate efficiently, and maintain
multiple versions of a project. Platforms such as GitHub and GitLab provide cloud-based repositories that support
team collaboration. Version control improves productivity and helps prevent conflicts during development.
"""

text4 = """
Once upon a time, in a small village surrounded by mountains, there lived a young boy who dreamed of becoming an
explorer. Every day he would climb the hills and imagine discovering new lands beyond the horizon. One day he found
an old map hidden inside a wooden chest. The map showed a mysterious island that no one in the village had ever
seen before. Determined to uncover its secrets, the boy prepared for an adventurous journey.
"""

text5 = """
A conversation between two colleagues during lunch discussed the increasing use of remote work. One colleague
explained that working from home improved productivity and reduced commuting stress. The other mentioned that
remote work sometimes leads to communication challenges and feelings of isolation. Both agreed that a hybrid work
model combining office collaboration and remote flexibility could be the best solution for modern workplaces.
"""

texts = [text1, text2, text3, text4, text5]

print("Dataset loaded with", len(texts), "texts")
```

```
Dataset loaded with 5 texts
```

STEP 4 — Generate Summaries

```
for i, text in enumerate(texts):
    # Tokenize the input text
    inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="pt")

    # Generate summary
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        min_length=25,
        max_length=60,
        early_stopping=True
    )

    # Decode the summary
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    print("\n----------------------------")
    print("Original Text", i+1)
    print(text)

    print("\nSummary", i+1)
    print(summary_text)
```

```
----------------------------
```

```
Original Text 1

Artificial Intelligence is transforming industries worldwide. Companies are increasingly adopting AI technologies
to automate processes, improve customer service, and enhance decision-making. Machine learning algorithms can
analyze massive amounts of data faster than humans, identifying patterns that were previously difficult to detect.
AI applications range from healthcare diagnostics to financial fraud detection and autonomous vehicles.
However, concerns regarding data privacy, ethical use of AI, and potential job displacement remain significant
challenges for policymakers and organizations.


Summary 1
Artificial Intelligence is transforming industries worldwide. Machine learning algorithms can identify patterns that were prev

----------------------------
Original Text 2

Climate change is one of the most critical challenges facing humanity today. Rising global temperatures are causing
melting glaciers, sea level rise, and extreme weather events. Scientists warn that without significant reductions
in greenhouse gas emissions, the effects will become more severe. Governments around the world are working toward
sustainable solutions such as renewable energy adoption, carbon reduction policies, and environmental conservation.
Public awareness and global cooperation are essential to mitigate climate risks.


Summary 2
Rising global temperatures are causing melting glaciers, sea level rise, and extreme weather events. Scientists warn that withc

----------------------------
Original Text 3

In software development, version control systems are essential tools for managing code changes. Git is one of the
most widely used systems that allows developers to track modifications, collaborate efficiently, and maintain
multiple versions of a project. Platforms such as GitHub and GitLab provide cloud-based repositories that support
team collaboration. Version control improves productivity and helps prevent conflicts during development.


Summary 3
Git is one of the most widely used systems that allows developers to track modifications. Platforms such as GitHub and GitLab p

----------------------------
Original Text 4

Once upon a time, in a small village surrounded by mountains, there lived a young boy who dreamed of becoming an
explorer. Every day he would climb the hills and imagine discovering new lands beyond the horizon. One day he found
an old map hidden inside a wooden chest. The map showed a mysterious island that no one in the village had ever
seen before. Determined to uncover its secrets, the boy prepared for an adventurous journey.


Summary 4
A young boy discovered a mysterious island in a wooden chest. He was determined to discover the island's secrets. The boy set c

----------------------------
Original Text 5

A conversation between two colleagues during lunch discussed the increasing use of remote work. One colleague
explained that working from home improved productivity and reduced commuting stress. The other mentioned that
```

STEP 5 — Control Summary Length

```
# Helper function to generate summaries with specific lengths
def generate_summary(text, min_len, max_len):
    inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="pt")
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        min_length=min_len,
        max_length=max_len,
        early_stopping=True
    )
    return tokenizer.decode(summary_ids[0], skip_special_tokens=True)

short_summary = generate_summary(text1, min_len=10, max_len=30)
print("Short Summary:")
print(short_summary)

long_summary = generate_summary(text1, min_len=50, max_len=120)
print("\nLong Summary:")
print(long_summary)

medium_summary = generate_summary(text1, min_len=25, max_len=60)
```

```
print("\nMedium Summary:")
print(medium_summary)
```

```
Short Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning algorithms can identify patterns that were previo

Long Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning algorithms can identify patterns that were previo

Medium Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning algorithms can identify patterns that were previo
```

STEP 6 — Compare Different Text Types

```
for i, text in enumerate(texts):
    # Tokenize the input text
    inputs = tokenizer(text, max_length=1024, truncation=True, return_tensors="pt")

    # Generate summary
    summary_ids = model.generate(
        inputs["input_ids"],
        num_beams=4,
        min_length=30,
        max_length=70,
        early_stopping=True
    )

    # Decode the summary
    summary_text = tokenizer.decode(summary_ids[0], skip_special_tokens=True)

    print("\n========== TEXT TYPE", i+1, "==========")
    print("Summary:")
    print(summary_text)
```

```
========== TEXT TYPE 1 ==========
Summary:
Artificial Intelligence is transforming industries worldwide. Machine learning algorithms can identify patterns that were previo

========== TEXT TYPE 2 ==========
Summary:
Rising global temperatures are causing melting glaciers, sea level rise, and extreme weather events. Scientists warn that withou

========== TEXT TYPE 3 ==========
Summary:
Git is one of the most widely used systems that allows developers to track modifications. Platforms such as GitHub and GitLab pr

========== TEXT TYPE 4 ==========
Summary:
A young boy discovered a mysterious island in a wooden chest. He was determined to discover the island's secrets. The boy set of

========== TEXT TYPE 5 ==========
Summary:
A conversation between two colleagues during lunch discussed the increasing use of remote work. One colleague explained that wor
```

STEP 7 — Test with Very Long Text

```
long_text = """
Artificial Intelligence has experienced rapid growth over the past decade. Researchers and companies are investing
billions of dollars into developing intelligent systems capable of performing complex tasks. AI technologies such
as deep learning, natural language processing, and computer vision have significantly improved the capabilities
of machines. In healthcare, AI assists doctors in diagnosing diseases earlier and more accurately. In transportation,
self-driving cars aim to reduce accidents and improve efficiency. Financial institutions use AI to detect fraudulent
transactions and manage risks.

Despite these advantages, AI also raises serious concerns. Ethical issues related to bias in algorithms, privacy
violations, and surveillance are increasingly debated. Governments and international organizations are attempting
to develop regulatory frameworks to ensure responsible AI deployment. Furthermore, automation driven by AI could
replace certain jobs, creating economic challenges for workers.

Experts believe the future of AI depends on balancing innovation with ethical considerations. Collaboration between
governments, researchers, and industries will be necessary to ensure that AI benefits society while minimizing
its potential harms.
"""

summary_long = generate_summary(long_text, min_len=40, max_len=100)
```

```
print("Summary of Long Text:")
print(summary_long)
```

```
Summary of Long Text:
Artificial Intelligence is being used to improve health care. But it is also raising concerns about privacy and the potential fo
```